

Neural Network and Deep Learning

Big Data Analytics

Readings

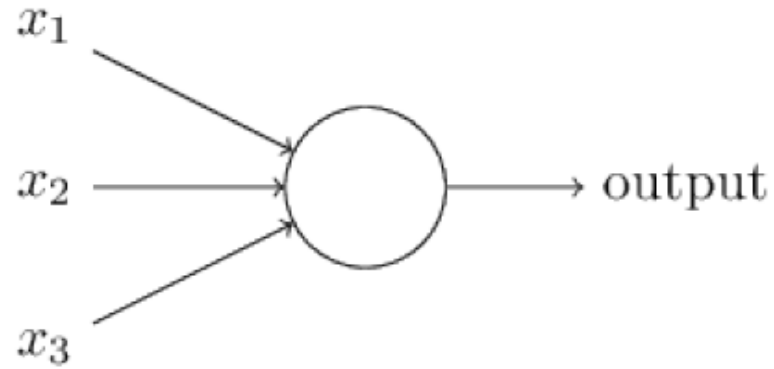
- Neural Network and Deep Learning
 - Chap. 1

Recap

Big Data Analytics

Sigmoid Neurons

- Sigmoid neurons look like Perceptrons



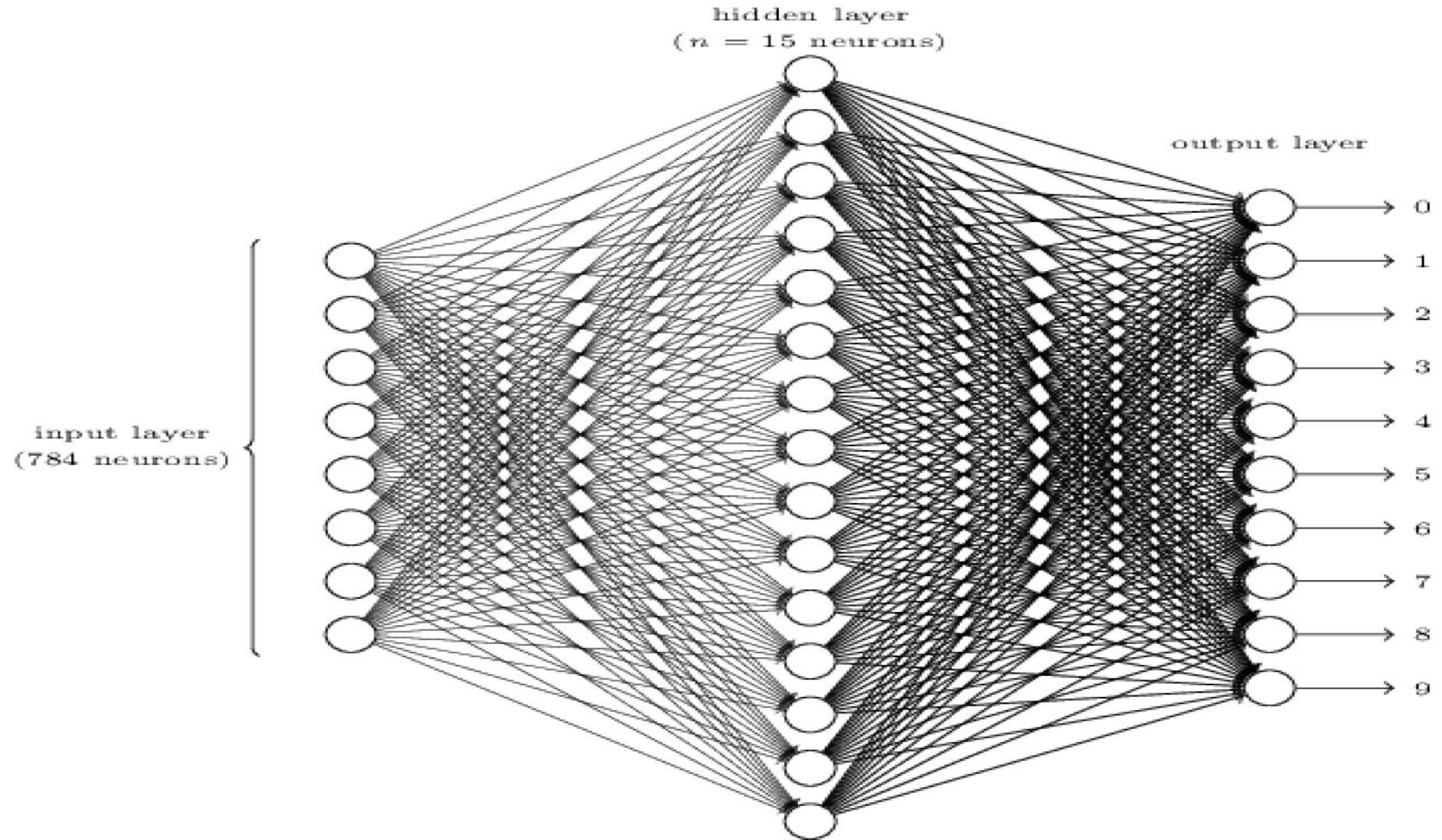
- Inputs can take on any values between 0 and 1
- Each input has weight
- Output is $\sigma(wx + b)$ where $\sigma(z) \equiv \frac{1}{1 + e^{-z}}$ $z = w \cdot x + b$

Using Neural Nets to Recognize Handwritten Digits



- MNIST dataset
 - 28 X 28 grey scale
 - 60,000 training set and 10,000 test set
 - 250 people wrote training set and another 250 people wrote test set

A simple network to classify handwritten digits



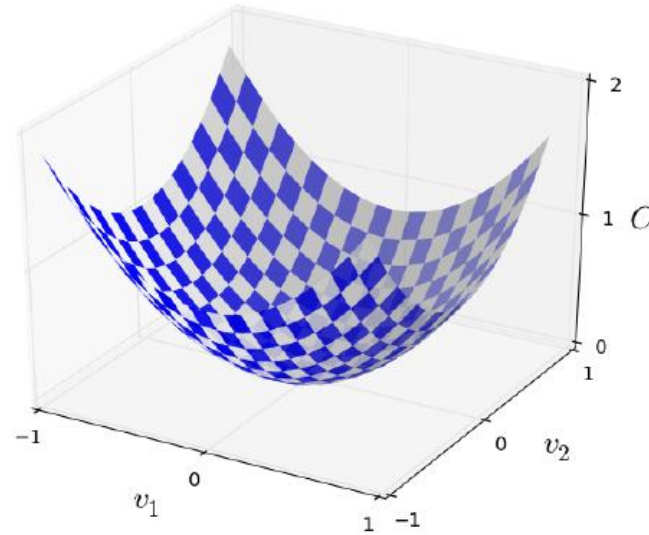
- Approximation of two variable function
- Min/Max problems
- Least Squares
- Gradient vector

Gradient descent

- Gradient descent is an iterative algorithm for finding the local minimum of a differentiable function
- Get a gradient vector of a cost function C , $\nabla C \equiv \left(\frac{\partial C}{\partial v_1}, \frac{\partial C}{\partial v_2} \right)^T$
- Decrease the input vector $\Delta v = -\eta \nabla C$, where η is a small positive parameter (learning rate)

Learning with gradient descent

- Let's suppose we're trying to minimize some function $C(v)$ where v is a vector $v_1, v_2, v_3, \dots, v_m$



- Then $\Delta C \approx \nabla C \cdot \Delta v$ where $\nabla C \equiv \left(\frac{\partial C}{\partial v_1}, \dots, \frac{\partial C}{\partial v_m} \right)^T$ and decrease $\Delta v = -\eta \nabla C$
$$v \rightarrow v' = v - \eta \nabla C$$

Learning with gradient descent

- Input x is a $28 \times 28 = 784$ -dimensional vector
- Desired output $y(x) = (0, 0, 0, 0, 0, 0, 1, 0, 0, 0)^T$ if a particular training image x , depicts a 6
- The cost function $C(w, b) \equiv \frac{1}{2n} \sum_x \|y(x) - a\|^2$ where
 - w denotes the collection of all weights in the network
 - b denotes all the biases
 - n is the total number of training inputs
 - a is the vector of outputs from the network when x is the input

Learning with gradient descent

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- Therefore $v \rightarrow v' = v - \eta \nabla C$

- Now for weight w_k and bias b_l

$$w_k \rightarrow w'_k = w_k - \eta \frac{\partial C}{\partial w_k}$$

$$b_l \rightarrow b'_l = b_l - \eta \frac{\partial C}{\partial b_l}.$$

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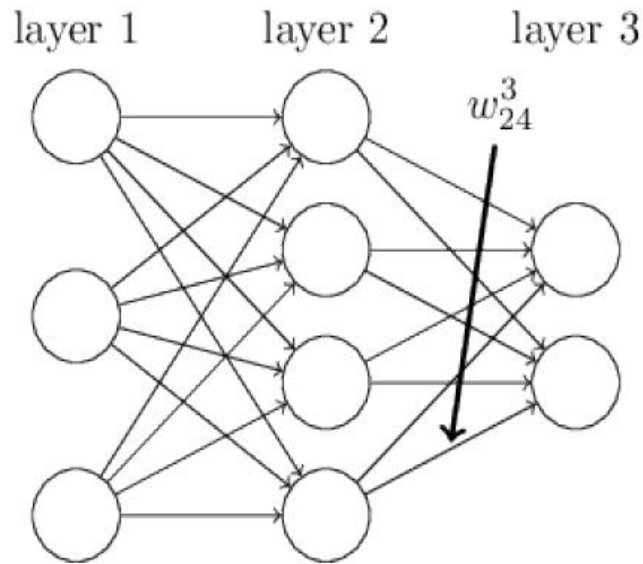
- Where $C(w, b) \equiv \frac{1}{2n} \sum_x \|y(x) - a\|^2$

$$C_x \equiv \frac{\|y(x) - a\|^2}{2} \text{ and } C = \frac{1}{n} \sum_x C_x$$

Matrix-based approach to compute the output

- The Notation

$$w_{jk}^l$$



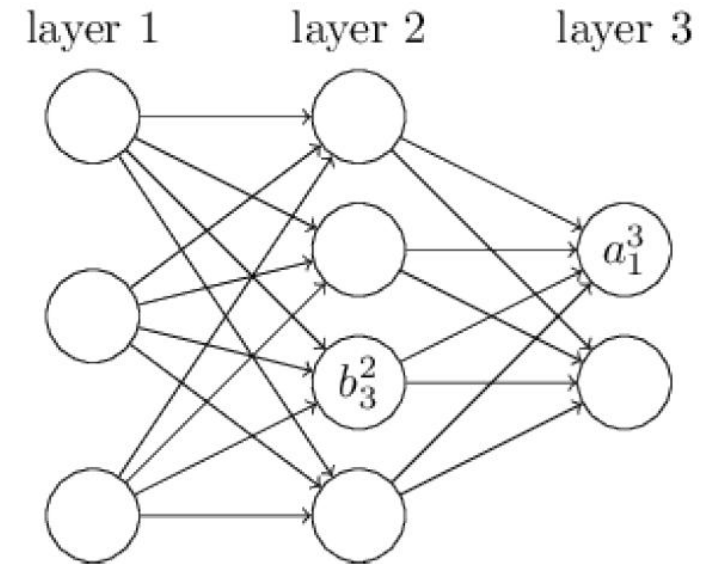
w_{jk}^l is the weight from the k^{th} neuron in the $(l-1)^{\text{th}}$ layer to the j^{th} neuron in the l^{th} layer

$$a_j^l = \sigma \left(\sum_k w_{jk}^l a_k^{l-1} + b_j^l \right)$$

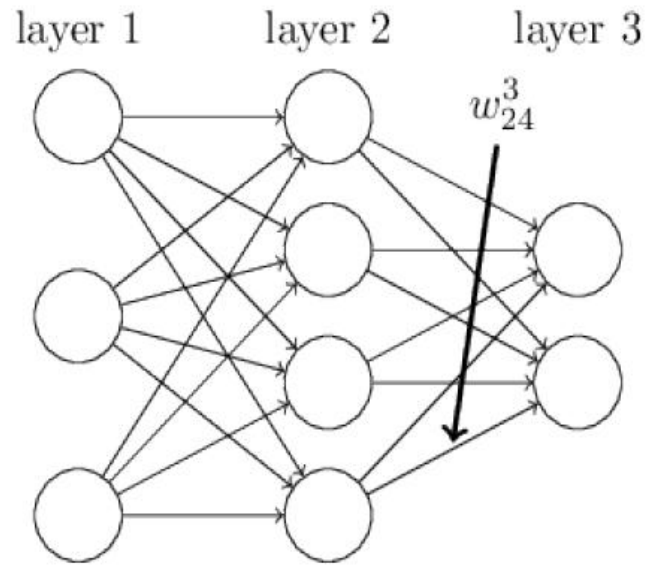
$$a^l = \sigma(w^l a^{l-1} + b^l)$$

$$a^l = \sigma(z^l)$$

$$b_j^l \quad a_j^l$$



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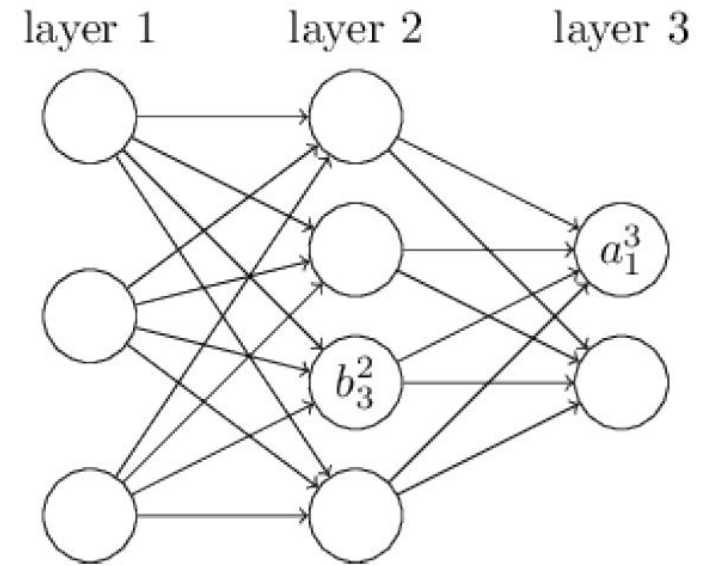
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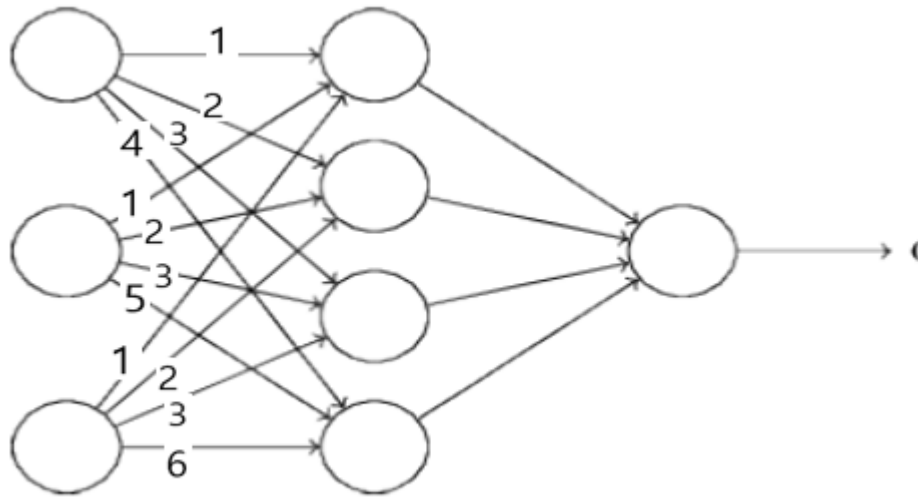
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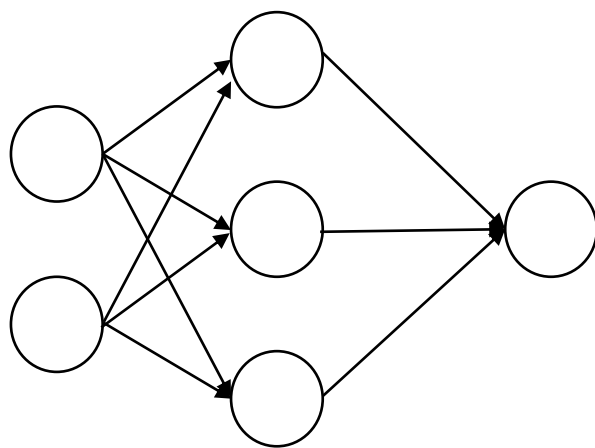




w^2 ?

z^2 ?

a^1 vector is $(0.1, 0.1, 0.7)^T$ **and** b^2 vector is $(3, 4, 5, 6)^T$



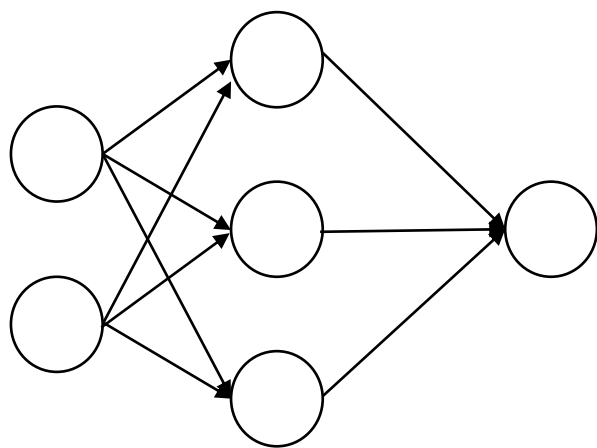
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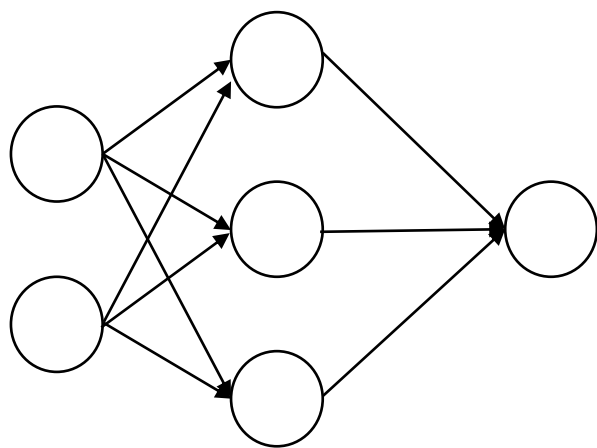
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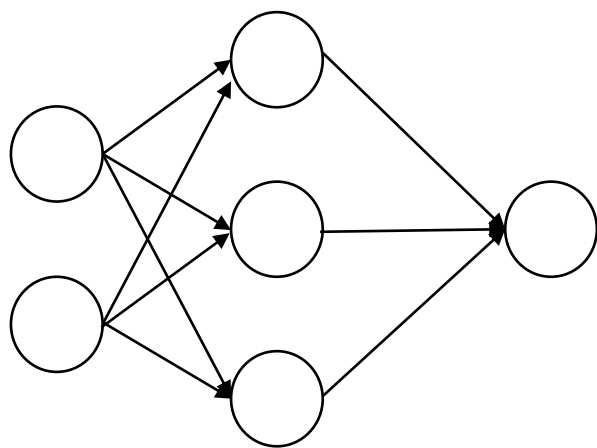
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Back Propagation

The Hadamard product, $s \odot t$

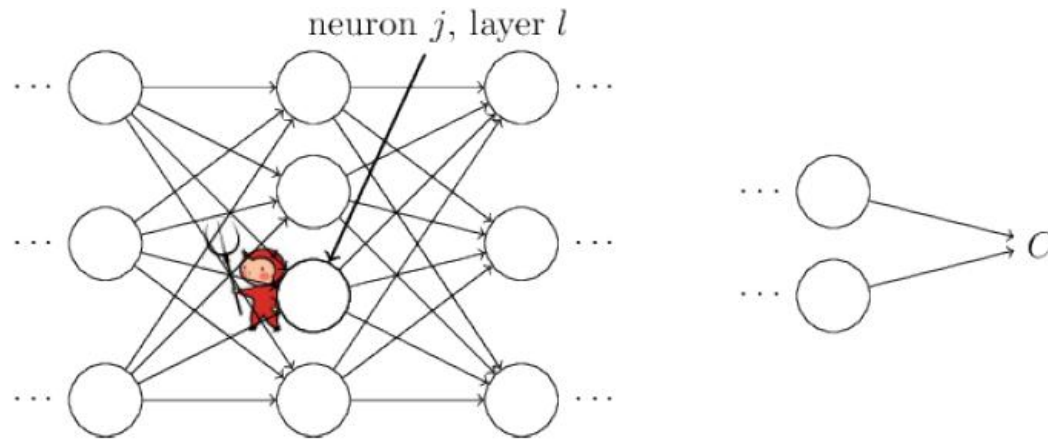
- Back Propagation algorithm is based on linear algebraic operations
 - Vector addition, multiplying a matrix by a vector, ...
- $s \odot t$: the elementwise product of the 2 vectors

$$\begin{bmatrix} 1 \\ 2 \end{bmatrix} \odot \begin{bmatrix} 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 1 * 3 \\ 2 * 4 \end{bmatrix} = \begin{bmatrix} 3 \\ 8 \end{bmatrix}$$

- Good matrix libraries usually provide fast implementation of the Hadamard product

Introducing the Error δ_j^l

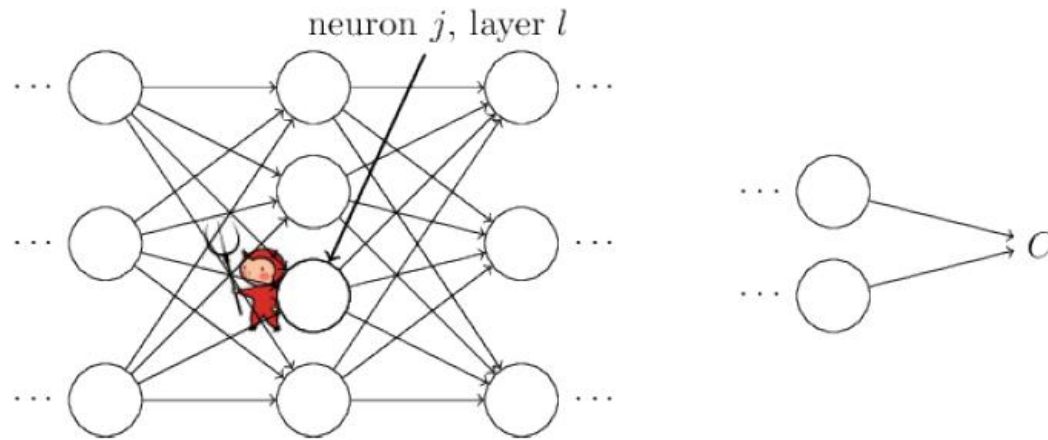
- δ_j^l is the error in the j -th neuron in the l -th layer
 - First compute δ_j^l and then relate δ_j^l to $\partial C / \partial w_{jk}^l$ and $\partial C / \partial b_j^l$



- Small change of z_j^l causes small change of C :
- The definition:

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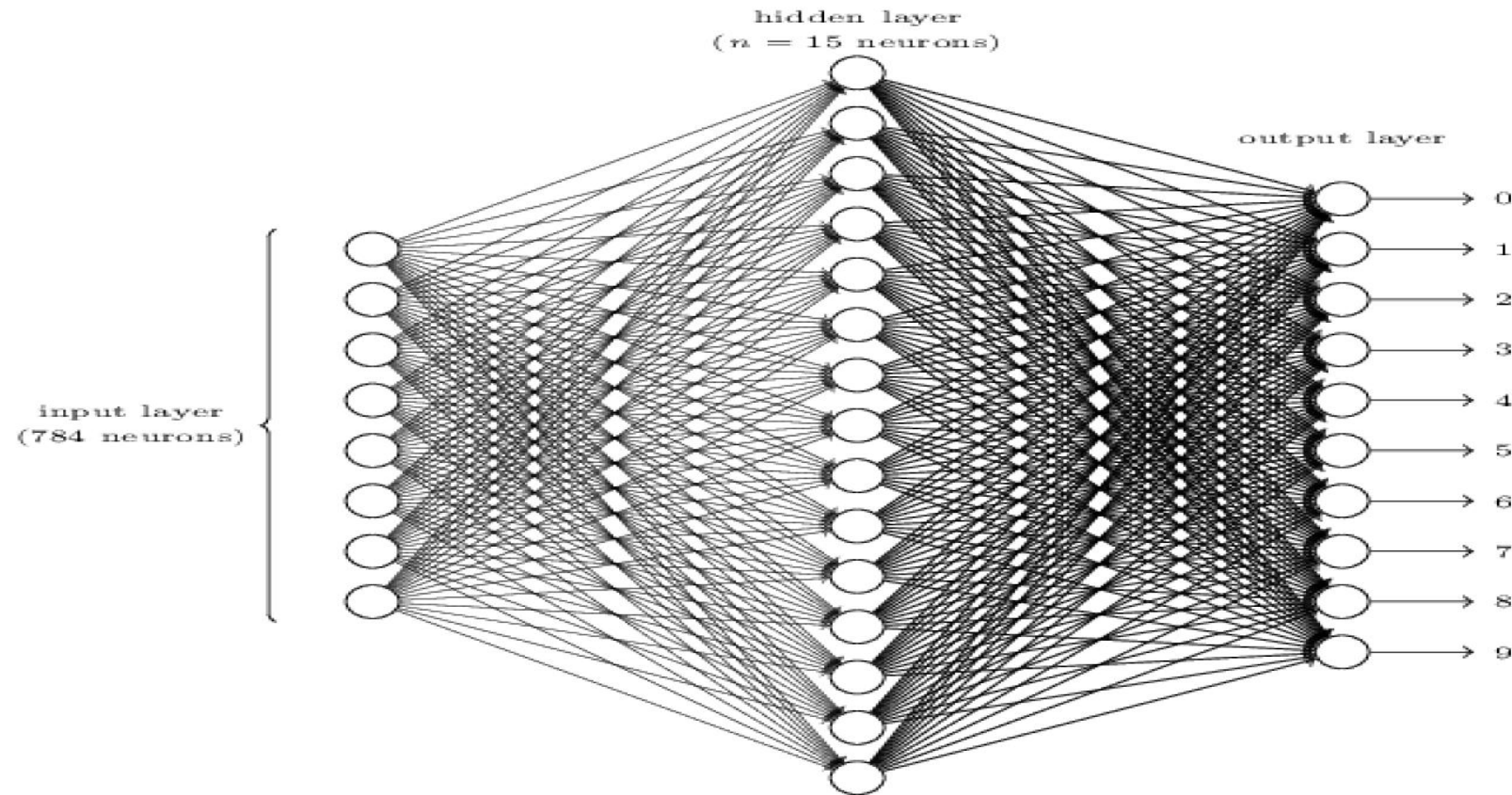
- Small change of z_j^l causes small change of C : $\frac{\partial C}{\partial z_j^l} \Delta z_j^l$
- The definition: $\delta_j^l \equiv \frac{\partial C}{\partial z_j^l}$

- BP1

$$\delta_j^L = \frac{\partial C}{\partial a_j^L} \sigma'(z_j^L)$$

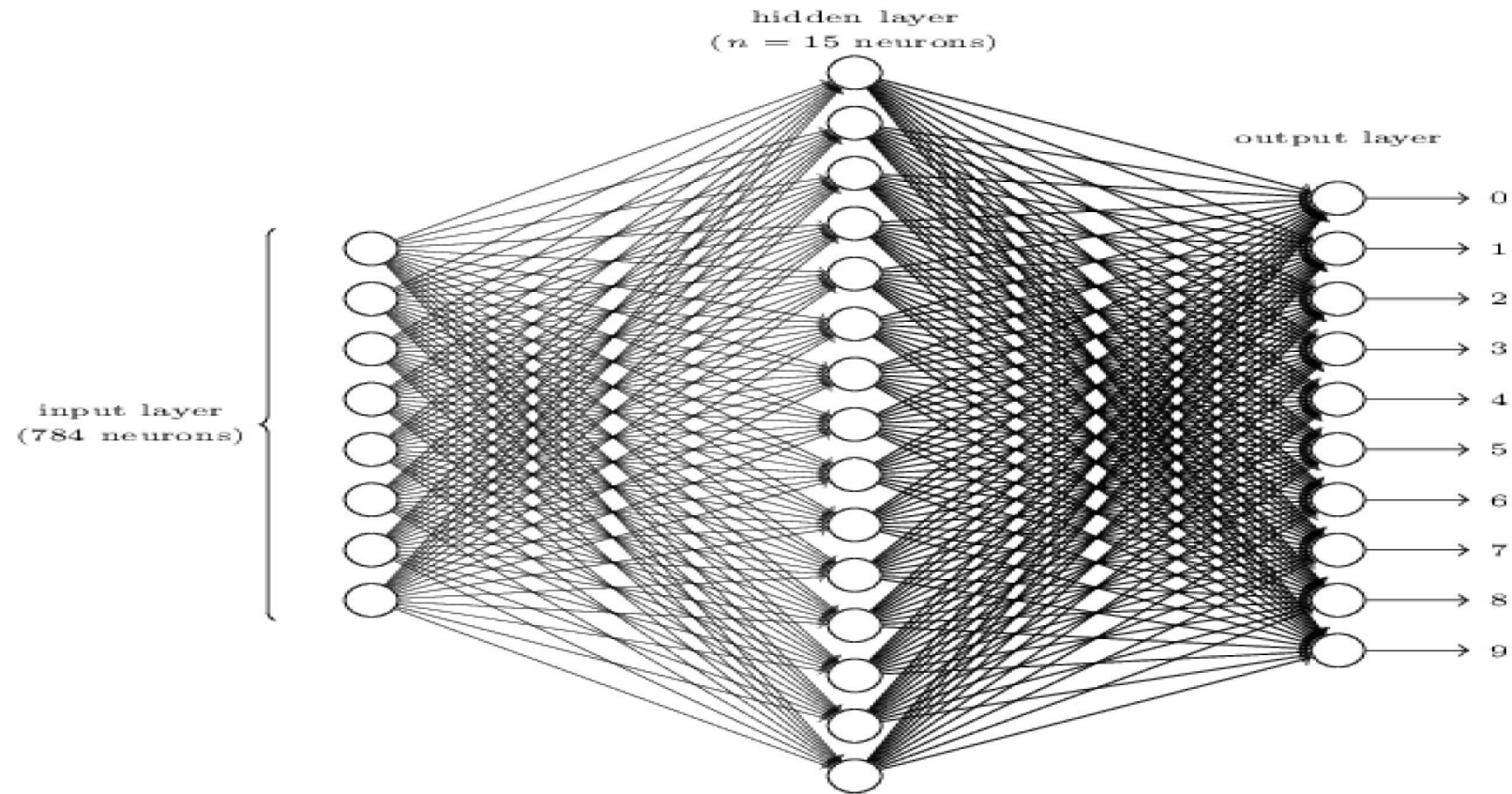
$$C = \frac{1}{2} \sum_j (y_j - a_j^L)^2$$

$$\partial C / \partial a_j^L = (a_j^L - y_j)$$



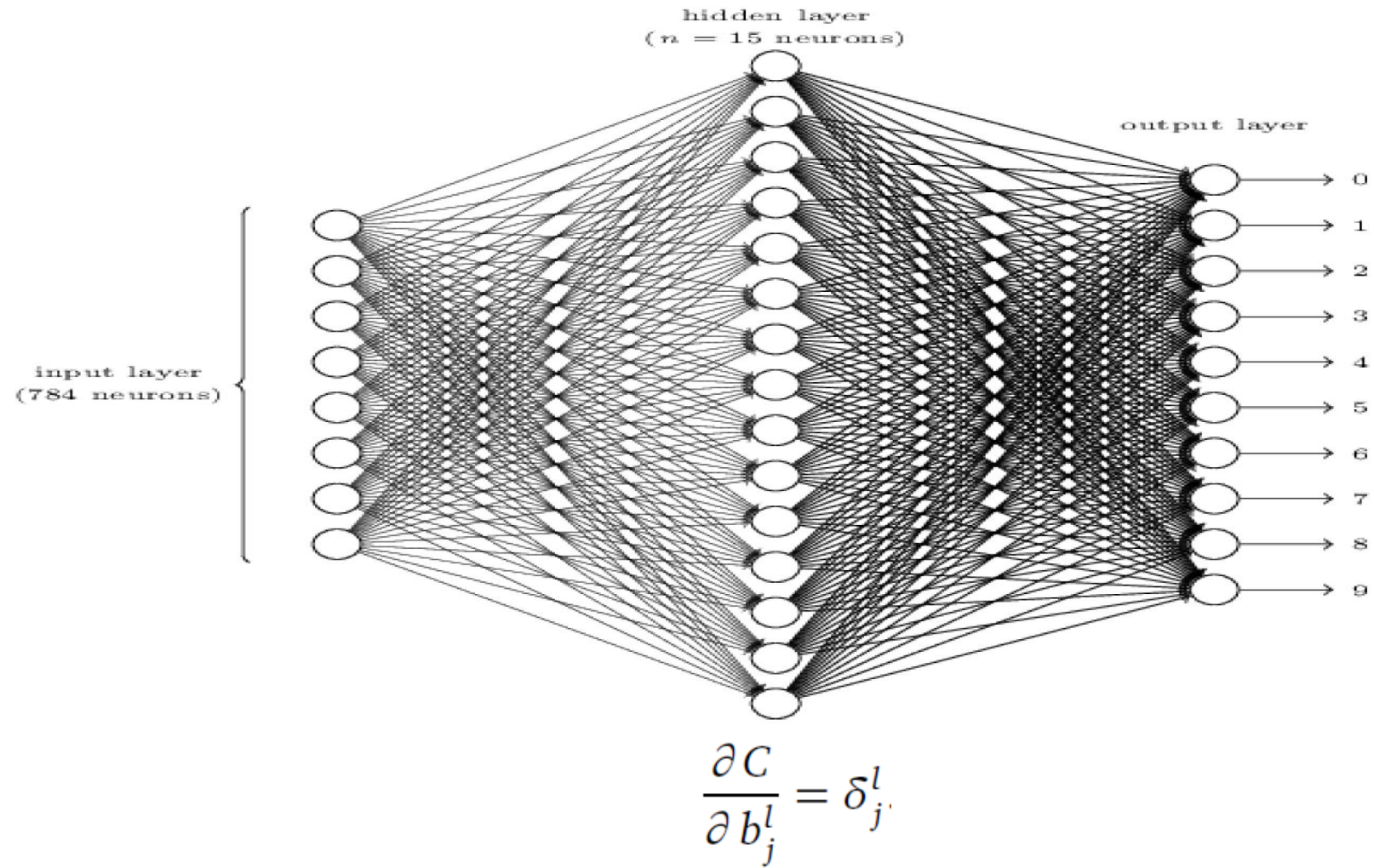
$$\delta^L = (a^L - y) \odot \sigma'(z^L)$$

- BP2

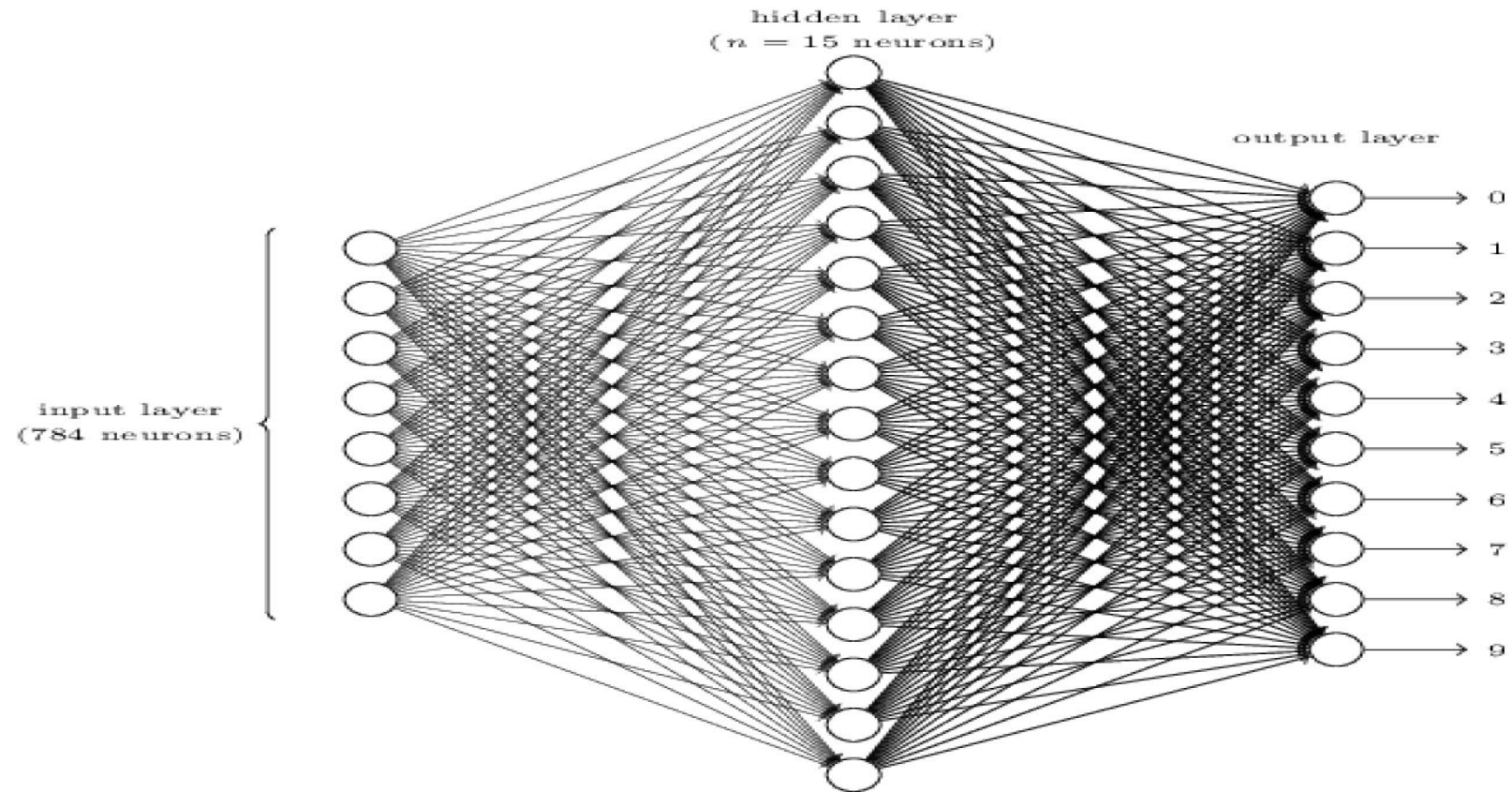


$$\delta^l = ((w^{l+1})^T \delta^{l+1}) \odot \sigma'(z^l)$$

- BP3



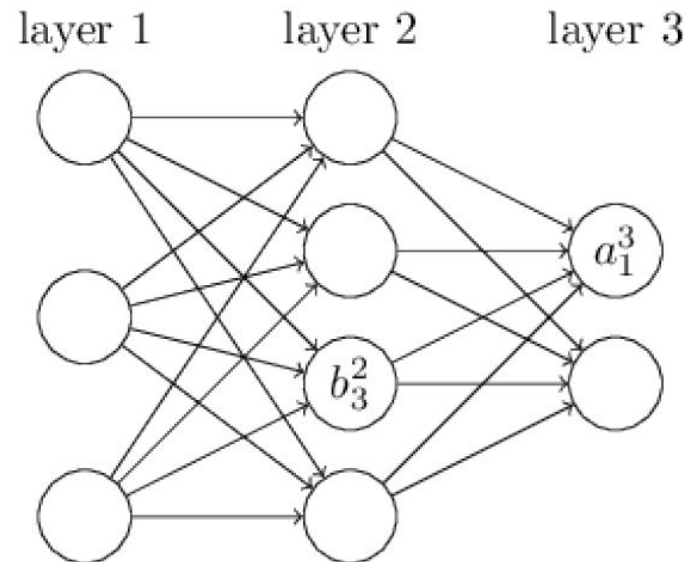
- BP4



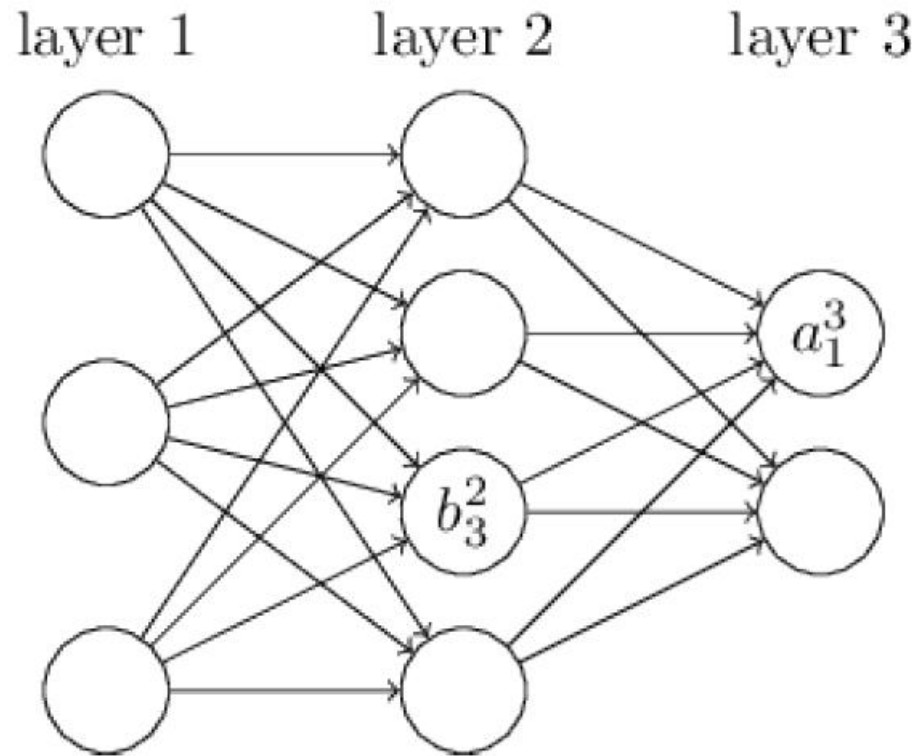
$$\frac{\partial C}{\partial w_{jk}^l} = a_k^{l-1} \delta_j^l$$

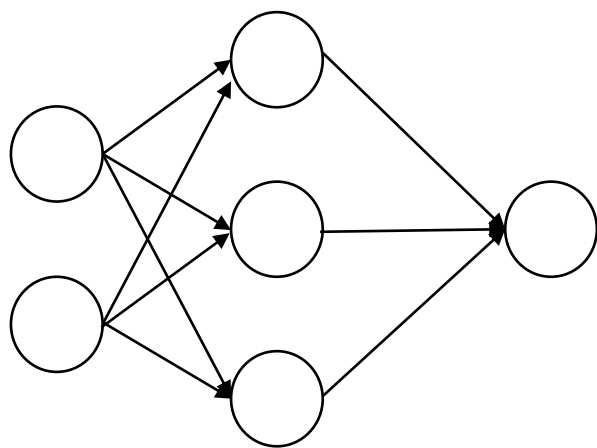
The Back Propagation Algorithm

1. **Input x :** Set the corresponding activation a^1 for the input layer.
2. **Feedforward:** For each $l = 2, 3, \dots, L$ compute $z^l = w^l a^{l-1} + b^l$ and $a^l = \sigma(z^l)$.
3. **Output error δ^L :** Compute the vector $\delta^L = \nabla_a C \odot \sigma'(z^L)$.
4. **Backpropagate the error:** For each $l = L-1, L-2, \dots, 2$ compute $\delta^l = ((w^{l+1})^T \delta^{l+1}) \odot \sigma'(z^l)$.
5. **Output:** The gradient of the cost function is given by $\frac{\partial C}{\partial w_{jk}^l} = a_k^{l-1} \delta_j^l$ and $\frac{\partial C}{\partial b_j^l} = \delta_j^l$.



The Back Propagation Algorithm





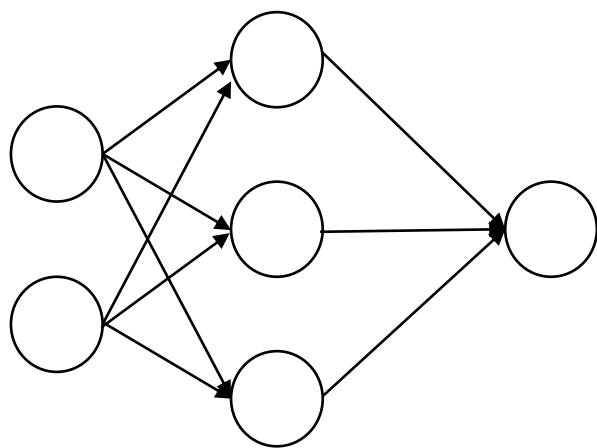
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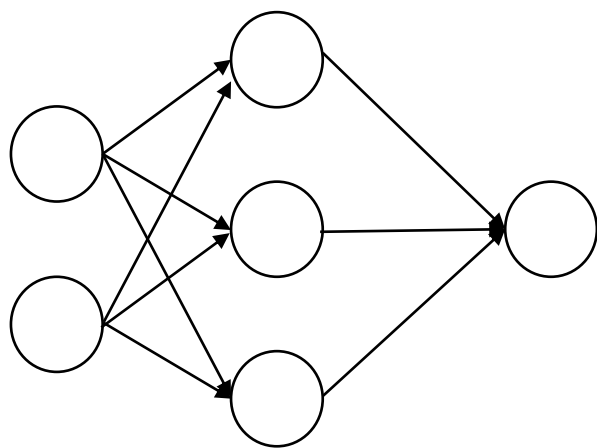
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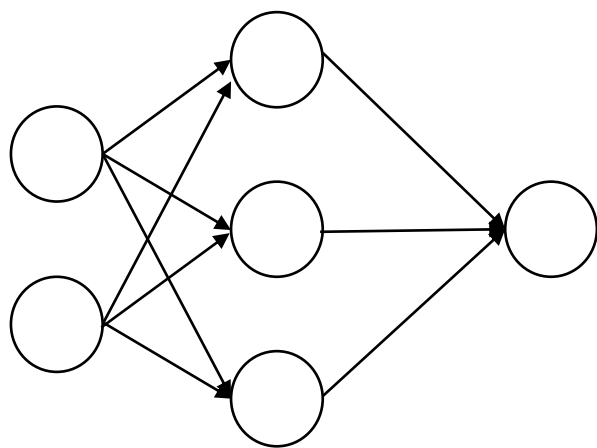
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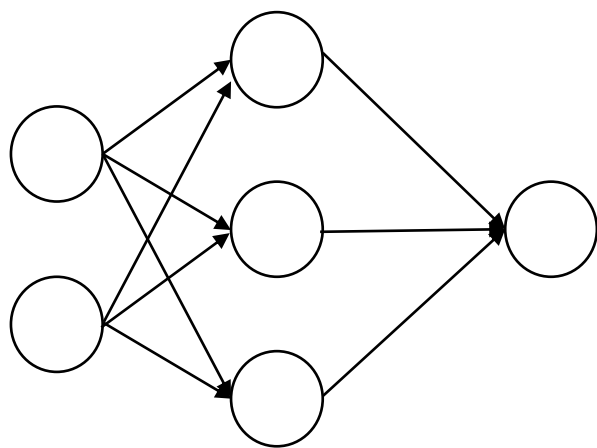
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• BP1 $\delta^L = (a^L - y) \odot \sigma'(z^L)$

If the label is 1

$$\delta^3 = [0.906 - 1] \odot \sigma'(2.262)$$

$$\begin{aligned} \sigma'(z) &= -1 \cdot (1 + e^{-z})^2 \cdot e^{-z} \cdot (-1) \\ &= \frac{e^{-z}}{(1 + e^{-z})^2} \end{aligned}$$

$$\sigma'(2.262) = [0.085] \quad \delta^3 = [-0.008]$$

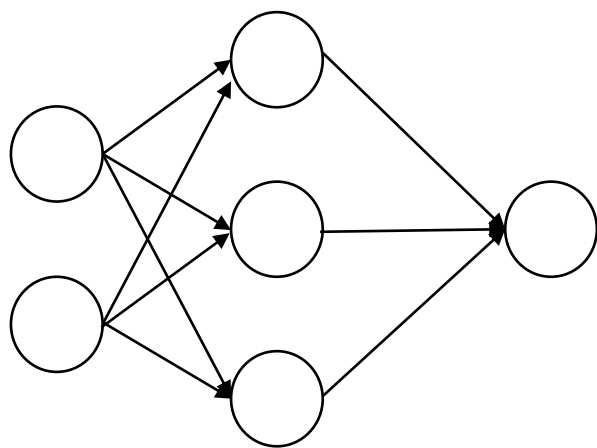
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If the label is 0

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$$\sigma'(2.262) = [0.085] \quad \delta^3 = [0.077]$$

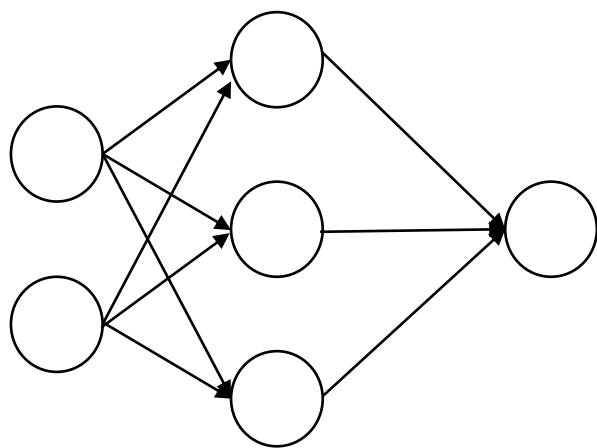
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if the label is 1
 $\delta^3 = [-0.008]$

if the label is 0
 $\delta^3 = [0.017]$

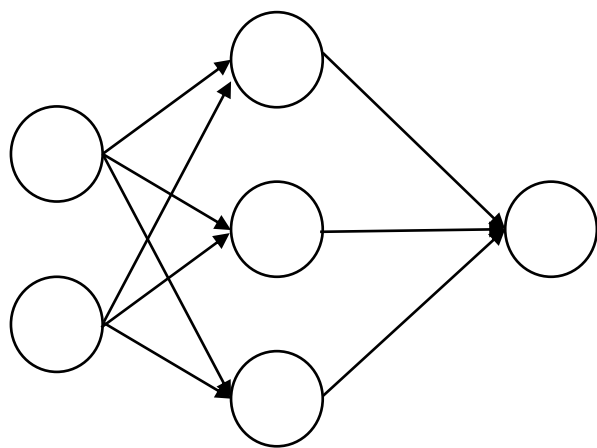
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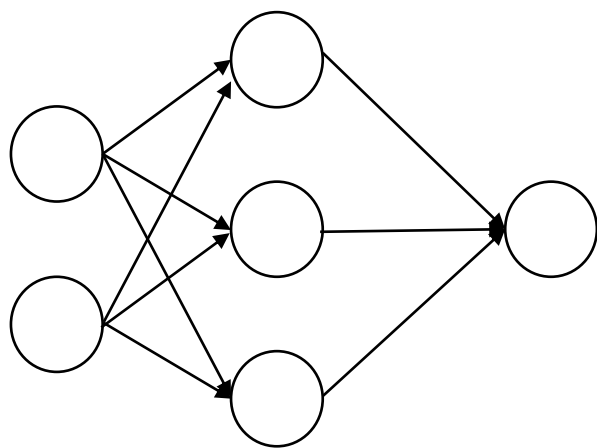
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 $\delta^3 = [0.017]$

• BP3 $\frac{\partial C}{\partial b_j^l} = \delta_j^l$ $\eta = 0.5$

$$b^{3'} = b^3 - \eta \delta^3 = [1.004] \text{ or } [0.9815]$$



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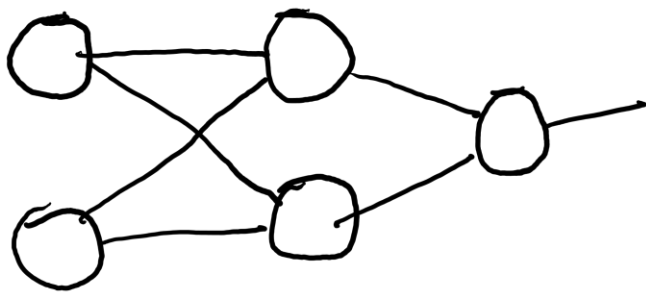
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• BP4 $\frac{\partial C}{\partial w_{jk}^l} = a_k^{l-1} \delta_j^l \quad \eta = 0.5$

$$\frac{\partial C}{\partial w_{11}^3} = -0.004 \quad \frac{\partial C}{\partial w_{12}^3} = -0.001 \quad \frac{\partial C}{\partial w_{13}^3} = -0.007$$

$$\frac{\partial C}{\partial w_{11}^3} = 0.039 \quad \frac{\partial C}{\partial w_{12}^3} = 0.009 \quad \frac{\partial C}{\partial w_{13}^3} = 0.068$$



$$a^1 = \begin{bmatrix} 0.5 \\ 0.5 \end{bmatrix} \quad y = [0]$$

$$W^2 = \begin{bmatrix} 1 & -1 \\ 1 & 0 \end{bmatrix}$$

$$b^2 = \begin{bmatrix} 0.1 \\ -0.1 \end{bmatrix}$$

$$W^3 = \begin{bmatrix} -1 & 1 \end{bmatrix}$$

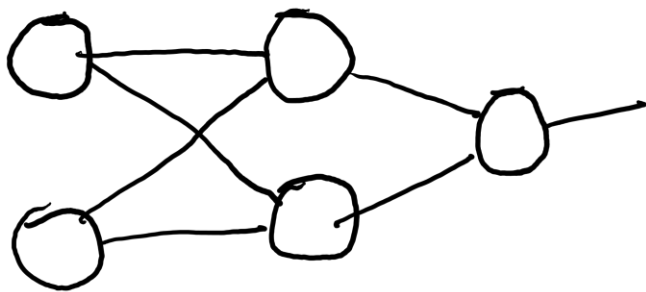
$$b^3 = [-0.5]$$

$$\delta^L = (a^L - y) \odot \sigma'(z^L)$$

$$\delta^l = ((w^{l+1})^T \delta^{l+1}) \odot \sigma'(z^l)$$

$$\frac{\partial C}{\partial b_i^l} = \delta_j^l$$

$$\frac{\partial C}{\partial w_{jk}^l} = a_k^{l-1} \delta_j^l$$



$$a^1 = \begin{bmatrix} 0.5 \\ 0.5 \end{bmatrix} \quad y = [0]$$

$$W^2 = \begin{bmatrix} 1 & -1 \\ 1 & 0 \end{bmatrix}$$

$$b^2 = \begin{bmatrix} 0.1 \\ -0.1 \end{bmatrix}$$

$$W^3 = \begin{bmatrix} -1 & 1 \end{bmatrix}$$

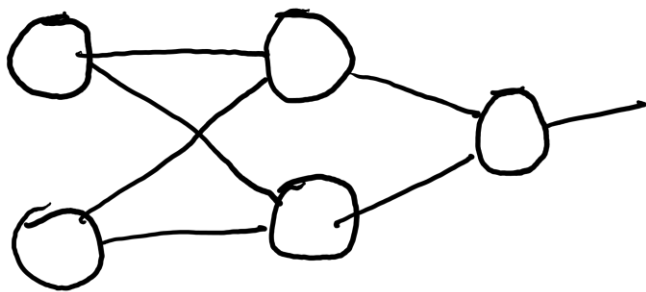
$$b^3 = [-0.5]$$

$$\delta^L = (a^L - y) \odot \sigma'(z^L)$$

$$\delta^l = ((w^{l+1})^T \delta^{l+1}) \odot \sigma'(z^l)$$

$$\frac{\partial C}{\partial b_i^l} = \delta_j^l$$

$$\frac{\partial C}{\partial w_{jk}^l} = a_k^{l-1} \delta_j^l$$



$$a^1 = \begin{bmatrix} 0.5 \\ 0.5 \end{bmatrix} \quad y = [0]$$

$$W^2 = \begin{bmatrix} 1 & -1 \\ 1 & 0 \end{bmatrix}$$

$$b^2 = \begin{bmatrix} 0.1 \\ -0.1 \end{bmatrix}$$

$$W^3 = \begin{bmatrix} -1 & 1 \end{bmatrix}$$

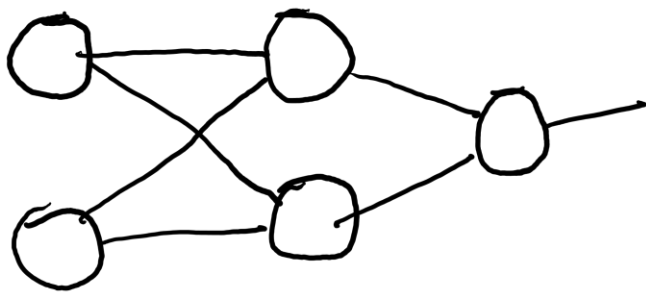
$$b^3 = [-0.5]$$

$$\delta^L = (a^L - y) \odot \sigma'(z^L)$$

$$\delta^l = ((w^{l+1})^T \delta^{l+1}) \odot \sigma'(z^l)$$

$$\frac{\partial C}{\partial b_i^l} = \delta_j^l$$

$$\frac{\partial C}{\partial w_{jk}^l} = a_k^{l-1} \delta_j^l$$



$$a^1 = \begin{bmatrix} 0.5 \\ 0.5 \end{bmatrix} \quad y = [0]$$

$$W^2 = \begin{bmatrix} 1 & -1 \\ 1 & 0 \end{bmatrix}$$

$$b^2 = \begin{bmatrix} 0.1 \\ -0.1 \end{bmatrix}$$

$$W^3 = \begin{bmatrix} -1 & 1 \end{bmatrix}$$

$$b^3 = [-0.5]$$

$$\delta^L = (a^L - y) \odot \sigma'(z^L)$$

$$\delta^l = ((w^{l+1})^T \delta^{l+1}) \odot \sigma'(z^l)$$

$$\frac{\partial C}{\partial b_i^l} = \delta_j^l$$

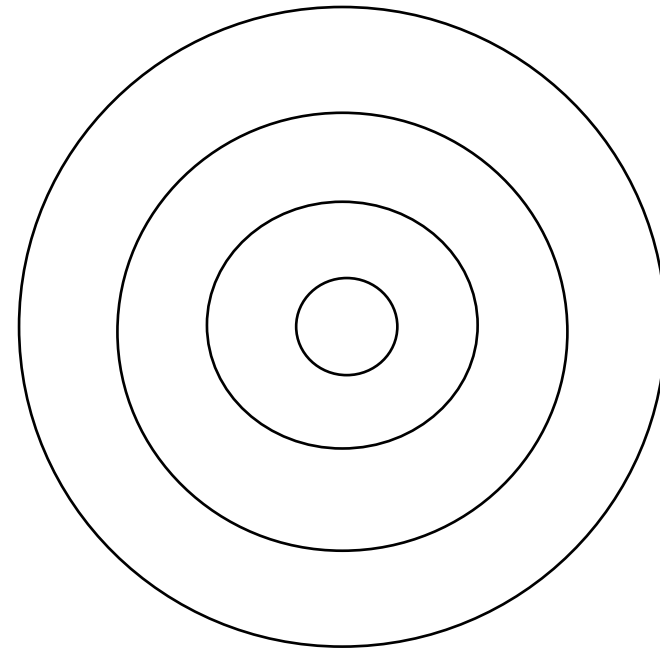
$$\frac{\partial C}{\partial w_{jk}^l} = a_k^{l-1} \delta_j^l$$

Stochastic Gradient Descent

- Estimate the gradient ∇C by computing ∇C_x for each sample of randomly chosen training inputs $X_1, X_2, \dots, X_n$:

$$w_k \rightarrow w'_k = w_k - \eta \sum_j \frac{\partial C_{X_j}}{\partial w_k}$$

$$b_l \rightarrow b'_l = b_l - \eta \sum_j \frac{\partial C_{X_j}}{\partial b_l}$$



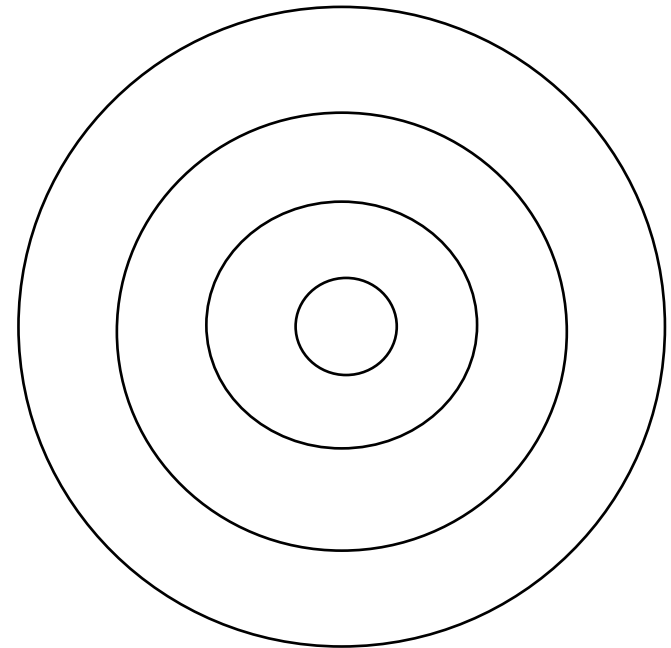
Mini Batch Gradient Descent

- Estimate the gradient ∇C by computing ∇C_{x_j} for a small sample of randomly chosen training inputs $X_1, X_2, \dots, X_m$:
 - Size m is large enough to expect that the average value of ∇C_{X_j} will be roughly equal to the average over all ∇C_x .

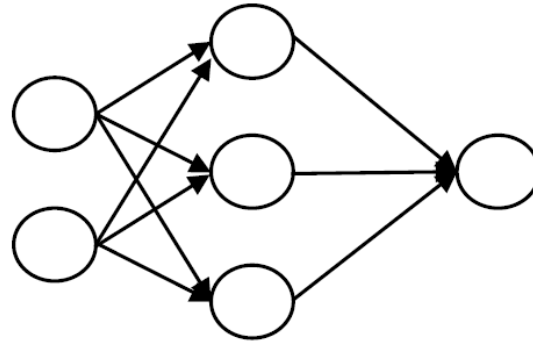
$$\frac{\sum_{j=1}^m \nabla C_{X_j}}{m} \approx \frac{\sum_x \nabla C_x}{n} = \nabla C \quad \nabla C \approx \frac{1}{m} \sum_{j=1}^m \nabla C_{X_j}$$

$$w_k \rightarrow w'_k = w_k - \frac{\eta}{m} \sum_j \frac{\partial C_{X_j}}{\partial w_k}$$

$$b_l \rightarrow b'_l = b_l - \frac{\eta}{m} \sum_j \frac{\partial C_{X_j}}{\partial b_l},$$



Below is a neural network:



All the matrices and vectors are as follows:

$$w^2 = \begin{bmatrix} 0.5 & -0.5 \\ 1 & 0 \\ -1 & 0.3 \end{bmatrix} w^3 = \begin{bmatrix} 1 & -1 & 0.5 \end{bmatrix} b^2 = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} b^3 = [0]$$

We are doing the mini-batch of size 4. And four inputs are

$$\mathbf{x}_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad \mathbf{x}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad \mathbf{x}_3 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \mathbf{x}_4 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Labels are

$$\mathbf{y}_1 = [0] \quad \mathbf{y}_2 = [1] \quad \mathbf{y}_3 = [1] \quad \mathbf{y}_4 = [0]$$

And the learning rate is $\eta = 0.4$

We are processing 4 inputs at the same time by having all 4 of them as a matrix.

Round off all the number to the nearest 1000th. For example, 6.6543 is 6.654 and 3.3355 is 3.336.

And the back propagation equations are as follows:

Summary: the equations of backpropagation

$$\delta^L = \nabla_a C \odot \sigma'(z^L) \quad (\text{BP1})$$

$$\delta^l = ((w^{l+1})^T \delta^{l+1}) \odot \sigma'(z^l) \quad (\text{BP2})$$

$$\frac{\partial C}{\partial b_j^l} = \delta_j^l \quad (\text{BP3})$$

$$\frac{\partial C}{\partial w_{jk}^l} = a_k^{l-1} \delta_j^l \quad (\text{BP4})$$

$$w^2 = \begin{bmatrix} 0.5 & -0.5 \\ 1 & 0 \\ -1 & 0.3 \end{bmatrix} \quad w^3 = \begin{bmatrix} 1 & -1 & 0.5 \end{bmatrix} \quad b^2 = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} \quad b^3 = \begin{bmatrix} 0 \end{bmatrix}$$

We are doing the mini-batch of size 4. And four inputs are

$$\mathbf{x}_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad \mathbf{x}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad \mathbf{x}_3 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \mathbf{x}_4 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Labels are

$$\mathbf{y}_1 = \begin{bmatrix} 0 \end{bmatrix} \quad \mathbf{y}_2 = \begin{bmatrix} 1 \end{bmatrix} \quad \mathbf{y}_3 = \begin{bmatrix} 1 \end{bmatrix} \quad \mathbf{y}_4 = \begin{bmatrix} 0 \end{bmatrix}$$

(a) [4 Points] What is Z^2 matrix? First represent it with matrix format and write the result. Show all your work.

And the back propagation equations are as follows:

Summary: the equations of backpropagation

$$\delta^L = \nabla_a C \odot \sigma'(z^L) \quad (\text{BP1})$$

$$\delta^l = ((w^{l+1})^T \delta^{l+1}) \odot \sigma'(z^l) \quad (\text{BP2})$$

$$\frac{\partial C}{\partial b_j^l} = \delta_j^l \quad (\text{BP3})$$

$$\frac{\partial C}{\partial w_{jk}^l} = a_k^{l-1} \delta_j^l \quad (\text{BP4})$$

$$w^2 = \begin{bmatrix} 0.5 & -0.5 \\ 1 & 0 \\ -1 & 0.3 \end{bmatrix} w^3 = \begin{bmatrix} 1 & -1 & 0.5 \end{bmatrix} b^2 = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} b^3 = \begin{bmatrix} 0 \end{bmatrix}$$

We are doing the mini-batch of size 4. And four inputs are

$$x_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad x_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad x_3 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad x_4 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Labels are

$$y_1 = \begin{bmatrix} 0 \end{bmatrix} \quad y_2 = \begin{bmatrix} 1 \end{bmatrix} \quad y_3 = \begin{bmatrix} 1 \end{bmatrix} \quad y_4 = \begin{bmatrix} 0 \end{bmatrix}$$

(a) [4 Points] What is Z^2 matrix? First represent it with matrix format and write the result. Show all your work.

$$Z^2 = \begin{bmatrix} 0.5 & -0.5 \\ 1 & 0 \\ -1 & 0.3 \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 1 & 1 & 1 \\ -1 & -1 & -1 & -1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0.5 & 1.5 & 1 \\ 0 & 0 & 1 & 1 \\ -1 & -0.7 & -2 & -1.7 \end{bmatrix}$$

(b) [4 Points] What is a^2 matrix? Show all your work.

$$w^2 = \begin{bmatrix} 0.5 & -0.5 \\ 1 & 0 \\ -1 & 0.3 \end{bmatrix} w^3 = \begin{bmatrix} 1 & -1 & 0.5 \end{bmatrix} b^2 = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} b^3 = \begin{bmatrix} 0 \end{bmatrix}$$

We are doing the mini-batch of size 4. And four inputs are

$$x_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad x_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad x_3 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad x_4 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Labels are

$$y_1 = \begin{bmatrix} 0 \end{bmatrix} \quad y_2 = \begin{bmatrix} 1 \end{bmatrix} \quad y_3 = \begin{bmatrix} 1 \end{bmatrix} \quad y_4 = \begin{bmatrix} 0 \end{bmatrix}$$

(c) [4 Points] What is Z^3 matrix? First represent it with matrix format and write the result. Show all your work.

(b) [4 Points] What is a^2 matrix? Show all your work.

$$w^2 = \begin{bmatrix} 0.5 & -0.5 \\ 1 & 0 \\ -1 & 0.3 \end{bmatrix} w^3 = \begin{bmatrix} 1 & -1 & 0.5 \end{bmatrix} b^2 = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} b^3 = [0]$$

$$a^2 = \begin{bmatrix} 0.731 & 0.622 & 0.818 & 0.731 \\ 0.5 & 0.5 & 0.731 & 0.731 \\ 0.269 & 0.332 & 0.119 & 0.154 \end{bmatrix}$$

We are doing the mini-batch of size 4. And four inputs are

$$x_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad x_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad x_3 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad x_4 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Labels are

$$y_1 = [0] \quad y_2 = [1] \quad y_3 = [1] \quad y_4 = [0]$$

(c) [4 Points] What is Z^3 matrix? First represent it with matrix format and write the result. Show all your work.

(b) [4 Points] What is a^2 matrix? Show all your work.

$$w^2 = \begin{bmatrix} 0.5 & -0.5 \\ 1 & 0 \\ -1 & 0.3 \end{bmatrix} w^3 = [1 \ -1 \ 0.5] b^2 = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} b^3 = [0]$$

$$a^2 = \begin{bmatrix} 0.731 & 0.622 & 0.818 & 0.731 \\ 0.5 & 0.5 & 0.731 & 0.731 \\ 0.269 & 0.332 & 0.119 & 0.154 \end{bmatrix}$$

We are doing the mini-batch of size 4. And four inputs are

$$x_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad x_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad x_3 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad x_4 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Labels are

$$y_1 = [0] \quad y_2 = [1] \quad y_3 = [1] \quad y_4 = [0]$$

(c) [4 Points] What is z^3 matrix? First represent it with matrix format and write the result. Show all your work.

$$\begin{aligned} z^3 &= [1 \ -1 \ 0.5] a^2 + [0 \ 0 \ 0 \ 0] \\ &= [0.366 \ 0.288 \ 0.147 \ 0.077] \end{aligned}$$

(d) [4 Points] What is a^3 matrix? Show all your work.

$$w^2 = \begin{bmatrix} 0.5 & -0.5 \\ 1 & 0 \\ -1 & 0.3 \end{bmatrix} w^3 = \begin{bmatrix} 1 & -1 & 0.5 \end{bmatrix} b^2 = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} b^3 = \begin{bmatrix} 0 \end{bmatrix}$$

We are doing the mini-batch of size 4. And four inputs are

$$x_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad x_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad x_3 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad x_4 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Labels are

$$y_1 = \begin{bmatrix} 0 \end{bmatrix} \quad y_2 = \begin{bmatrix} 1 \end{bmatrix} \quad y_3 = \begin{bmatrix} 1 \end{bmatrix} \quad y_4 = \begin{bmatrix} 0 \end{bmatrix}$$

(e) [3 Points] What is ${}^1\delta^3$ value? ${}^1\delta^3$ is the layer 3 delta for input 1. Show all your work.

(d) [4 Points] What is a^3 matrix? Show all your work.

$$w^2 = \begin{bmatrix} 0.5 & -0.5 \\ 1 & 0 \\ -1 & 0.3 \end{bmatrix} w^3 = \begin{bmatrix} 1 & -1 & 0.5 \end{bmatrix} b^2 = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} b^3 = \begin{bmatrix} 0 \end{bmatrix}$$

We are doing the mini-batch of size 4. And four inputs are

$$x_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad x_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad x_3 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad x_4 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Labels are

$$y_1 = \begin{bmatrix} 0 \end{bmatrix} \quad y_2 = \begin{bmatrix} 1 \end{bmatrix} \quad y_3 = \begin{bmatrix} 1 \end{bmatrix} \quad y_4 = \begin{bmatrix} 0 \end{bmatrix}$$

$$a^3 = \begin{bmatrix} 0.590 & 0.572 & 0.537 & 0.519 \end{bmatrix}$$

(e) [3 Points] What is ${}^1\delta^3$ value? ${}^1\delta^3$ is the layer 3 delta for input 1. Show all your work.

(d) [4 Points] What is a^3 matrix? Show all your work.

$$w^2 = \begin{bmatrix} 0.5 & -0.5 \\ 1 & 0 \\ -1 & 0.3 \end{bmatrix} w^3 = \begin{bmatrix} 1 & -1 & 0.5 \end{bmatrix} b^2 = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} b^3 = \begin{bmatrix} 0 \end{bmatrix}$$

We are doing the mini-batch of size 4. And four inputs are

$$x_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad x_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad x_3 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad x_4 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Labels are

$$y_1 = \begin{bmatrix} 0 \end{bmatrix} \quad y_2 = \begin{bmatrix} 1 \end{bmatrix} \quad y_3 = \begin{bmatrix} 1 \end{bmatrix} \quad y_4 = \begin{bmatrix} 0 \end{bmatrix}$$

$$a^3 = \begin{bmatrix} 0.590 & 0.572 & 0.537 & 0.519 \end{bmatrix}$$

(e) [3 Points] What is ${}^1\delta^3$ value? ${}^1\delta^3$ is the layer 3 delta for input 1. Show all your work.

$${}^1\delta^3 = 0.590 \times 0.590 \times (1 - 0.590) = 0.143$$

(f) [3 Points] What is ${}^2\delta^3$ value? ${}^2\delta^3$ is the layer 3 delta for input 2. Show all your work.

$$w^2 = \begin{bmatrix} 0.5 & -0.5 \\ 1 & 0 \\ -1 & 0.3 \end{bmatrix} w^3 = \begin{bmatrix} 1 & -1 & 0.5 \end{bmatrix} b^2 = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} b^3 = \begin{bmatrix} 0 \end{bmatrix}$$

We are doing the mini-batch of size 4. And four inputs are

$$x_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad x_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad x_3 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad x_4 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Labels are

$$y_1 = \begin{bmatrix} 0 \end{bmatrix} \quad y_2 = \begin{bmatrix} 1 \end{bmatrix} \quad y_3 = \begin{bmatrix} 1 \end{bmatrix} \quad y_4 = \begin{bmatrix} 0 \end{bmatrix}$$

(g) [3 Points] What is ${}^3\delta^3$ value? ${}^3\delta^3$ is the layer 3 delta for input 3. Show all your work.

(f) [3 Points] What is ${}^2\delta^3$ value? ${}^2\delta^3$ is the layer 3 delta for input 2. Show all your work.

$$w^2 = \begin{bmatrix} 0.5 & -0.5 \\ 1 & 0 \\ -1 & 0.3 \end{bmatrix} w^3 = \begin{bmatrix} 1 & -1 & 0.5 \end{bmatrix} b^2 = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} b^3 = \begin{bmatrix} 0 \end{bmatrix}$$

We are doing the mini-batch of size 4. And four inputs are

$$x_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad x_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad x_3 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad x_4 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Labels are

$$y_1 = \begin{bmatrix} 0 \end{bmatrix} \quad y_2 = \begin{bmatrix} 1 \end{bmatrix} \quad y_3 = \begin{bmatrix} 1 \end{bmatrix} \quad y_4 = \begin{bmatrix} 0 \end{bmatrix}$$

$${}^2\delta^3 = (0.572 - 1) \times 0.572 \times (1 - 0.572) = -0.105$$

(g) [3 Points] What is ${}^3\delta^3$ value? ${}^3\delta^3$ is the layer 3 delta for input 3. Show all your work.

(f) [3 Points] What is ${}^2\delta^3$ value? ${}^2\delta^3$ is the layer 3 delta for input 2. Show all your work.

$$w^2 = \begin{bmatrix} 0.5 & -0.5 \\ 1 & 0 \\ -1 & 0.3 \end{bmatrix} w^3 = \begin{bmatrix} 1 & -1 & 0.5 \end{bmatrix} b^2 = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} b^3 = \begin{bmatrix} 0 \end{bmatrix}$$

We are doing the mini-batch of size 4. And four inputs are

$$x_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad x_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad x_3 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad x_4 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Labels are

$$y_1 = \begin{bmatrix} 0 \end{bmatrix} \quad y_2 = \begin{bmatrix} 1 \end{bmatrix} \quad y_3 = \begin{bmatrix} 1 \end{bmatrix} \quad y_4 = \begin{bmatrix} 0 \end{bmatrix}$$

$${}^2\delta^3 = (0.572 - 1) \times 0.572 \times (1 - 0.572) = -0.105$$

(g) [3 Points] What is ${}^3\delta^3$ value? ${}^3\delta^3$ is the layer 3 delta for input 3. Show all your work.

$${}^3\delta^3 = (0.537 - 1) \times 0.537 \times (1 - 0.537) = -0.115$$

(h) [3 Points] What is δ^3 value? δ^3 is the layer 3 delta for input 4. Show all your work.

$$w^2 = \begin{bmatrix} 0.5 & -0.5 \\ 1 & 0 \\ -1 & 0.3 \end{bmatrix} w^3 = \begin{bmatrix} 1 & -1 & 0.5 \end{bmatrix} b^2 = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} b^3 = \begin{bmatrix} 0 \end{bmatrix}$$

We are doing the mini-batch of size 4. And four inputs are

$$x_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad x_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad x_3 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad x_4 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Labels are

$$y_1 = \begin{bmatrix} 0 \end{bmatrix} \quad y_2 = \begin{bmatrix} 1 \end{bmatrix} \quad y_3 = \begin{bmatrix} 1 \end{bmatrix} \quad y_4 = \begin{bmatrix} 0 \end{bmatrix}$$

(i) [4 Points] What is average $\frac{\partial C}{\partial b^3}$ vector? Show all your work.

(h) [3 Points] What is δ^3 value? δ^3 is the layer 3 delta for input 4. Show all your work.

$$w^2 = \begin{bmatrix} 0.5 & -0.5 \\ 1 & 0 \\ -1 & 0.3 \end{bmatrix} w^3 = \begin{bmatrix} 1 & -1 & 0.5 \end{bmatrix} b^2 = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} b^3 = [0]$$

We are doing the mini-batch of size 4. And four inputs are

$$x_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad x_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad x_3 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad x_4 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Labels are

$$y_1 = [0] \quad y_2 = [1] \quad y_3 = [1] \quad y_4 = [0]$$

$$\delta^3 = 0.519 \times 0.519 \times (1 - 0.519) = 0.130$$

(i) [4 Points] What is average $\frac{\partial C}{\partial b^3}$ vector? Show all your work.

(h) [3 Points] What is δ^3 value? δ^3 is the layer 3 delta for input 4. Show all your work.

$$w^2 = \begin{bmatrix} 0.5 & -0.5 \\ 1 & 0 \\ -1 & 0.3 \end{bmatrix} w^3 = [1 \ -1 \ 0.5] b^2 = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} b^3 = [0]$$

We are doing the mini-batch of size 4. And four inputs are

$$x_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad x_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad x_3 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad x_4 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Labels are

$$y_1 = [0] \quad y_2 = [1] \quad y_3 = [1] \quad y_4 = [0]$$

$$\delta^3 = 0.519 \times 0.519 \times (1 - 0.519) = 0.130$$

(i) [4 Points] What is average $\frac{\partial C}{\partial b^3}$ vector? Show all your work.

$$\begin{aligned} \frac{\partial C}{\partial b^3} &= [(0.143 - 0.105 - 0.115 + 0.130)/4] \\ &= [0.013] \end{aligned}$$

(j) [12 Points] What is average $\frac{\partial C}{\partial w^3}$ matrix? Show all your work.

$$w^2 = \begin{bmatrix} 0.5 & -0.5 \\ 1 & 0 \\ -1 & 0.3 \end{bmatrix} w^3 = \begin{bmatrix} 1 & -1 & 0.5 \end{bmatrix} b^2 = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} b^3 = \begin{bmatrix} 0 \end{bmatrix}$$

We are doing the mini-batch of size 4. And four inputs are

$$x_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad x_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad x_3 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad x_4 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Labels are

$$y_1 = \begin{bmatrix} 0 \end{bmatrix} \quad y_2 = \begin{bmatrix} 1 \end{bmatrix} \quad y_3 = \begin{bmatrix} 1 \end{bmatrix} \quad y_4 = \begin{bmatrix} 0 \end{bmatrix}$$

(j) [12 Points] What is average $\frac{\partial C}{\partial w^3}$ matrix? Show all your work.

$$w^2 = \begin{bmatrix} 0.5 & -0.5 \\ 1 & 0 \\ -1 & 0.3 \end{bmatrix} w^3 = \begin{bmatrix} 1 & -1 & 0.5 \end{bmatrix} b^2 = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} b^3 = \begin{bmatrix} 0 \end{bmatrix}$$

We are doing the mini-batch of size 4. And four inputs are

$$x_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad x_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad x_3 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad x_4 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Labels are

$$y_1 = \begin{bmatrix} 0 \end{bmatrix} \quad y_2 = \begin{bmatrix} 1 \end{bmatrix} \quad y_3 = \begin{bmatrix} 1 \end{bmatrix} \quad y_4 = \begin{bmatrix} 0 \end{bmatrix}$$

$$\begin{aligned} \frac{\partial C}{\partial w_{11}^3} &= (0.143 \times 0.731 - 0.105 \times 0.622 - 0.115 \times 0.818 + 0.130 \times 0.731) / 4 \\ &= 0.010 \end{aligned}$$

$$\begin{aligned} \frac{\partial C}{\partial w_{12}^3} &= (0.143 \times 0.5 - 0.105 \times 0.5 - 0.115 \times 0.731 + 0.130 \times 0.731) / 4 \\ &= 0.007 \end{aligned}$$

$$\begin{aligned} \frac{\partial C}{\partial w_{13}^3} &= (0.143 \times 0.269 - 0.105 \times 0.332 - 0.115 \times 0.119 + 0.130 \times 0.154) / 4 \\ &= 0.002 \end{aligned}$$

$$\frac{\partial C}{\partial w^3} = \begin{bmatrix} 0.010 & 0.007 & 0.002 \end{bmatrix}$$

$$w^2 = \begin{bmatrix} 0.5 & -0.5 \\ 1 & 0 \\ -1 & 0.3 \end{bmatrix} w^3 = \begin{bmatrix} 1 & -1 & 0.5 \end{bmatrix} b^2 = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} b^3 = \begin{bmatrix} 0 \end{bmatrix}$$

We are doing the mini-batch of size 4. And four inputs are

$$x_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad x_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad x_3 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad x_4 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Labels are

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(k) [3 Points] What is new b^3 vector? Show all your work.

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(k) [3 Points] What is new b^3 vector? Show all your work.

$$b^3 = [0] - 0.4 [0.0 \ 3] = [-0.005]$$

(I) [3 Points] What is new w^3 matrix? Show all your work.

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$$\begin{aligned} w^3 &= [1 \quad -1 \quad 0.5] - 0.4 \times [0.010 \quad 0.007 \quad 0.002] \\ &= [0.996 \quad -1.003 \quad 0.499] \end{aligned}$$

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if $i = j$ and $h = k$

$$\frac{\partial z_j^l}{\partial w_{jk}^{l-1}} = a_k^{l-1}$$

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